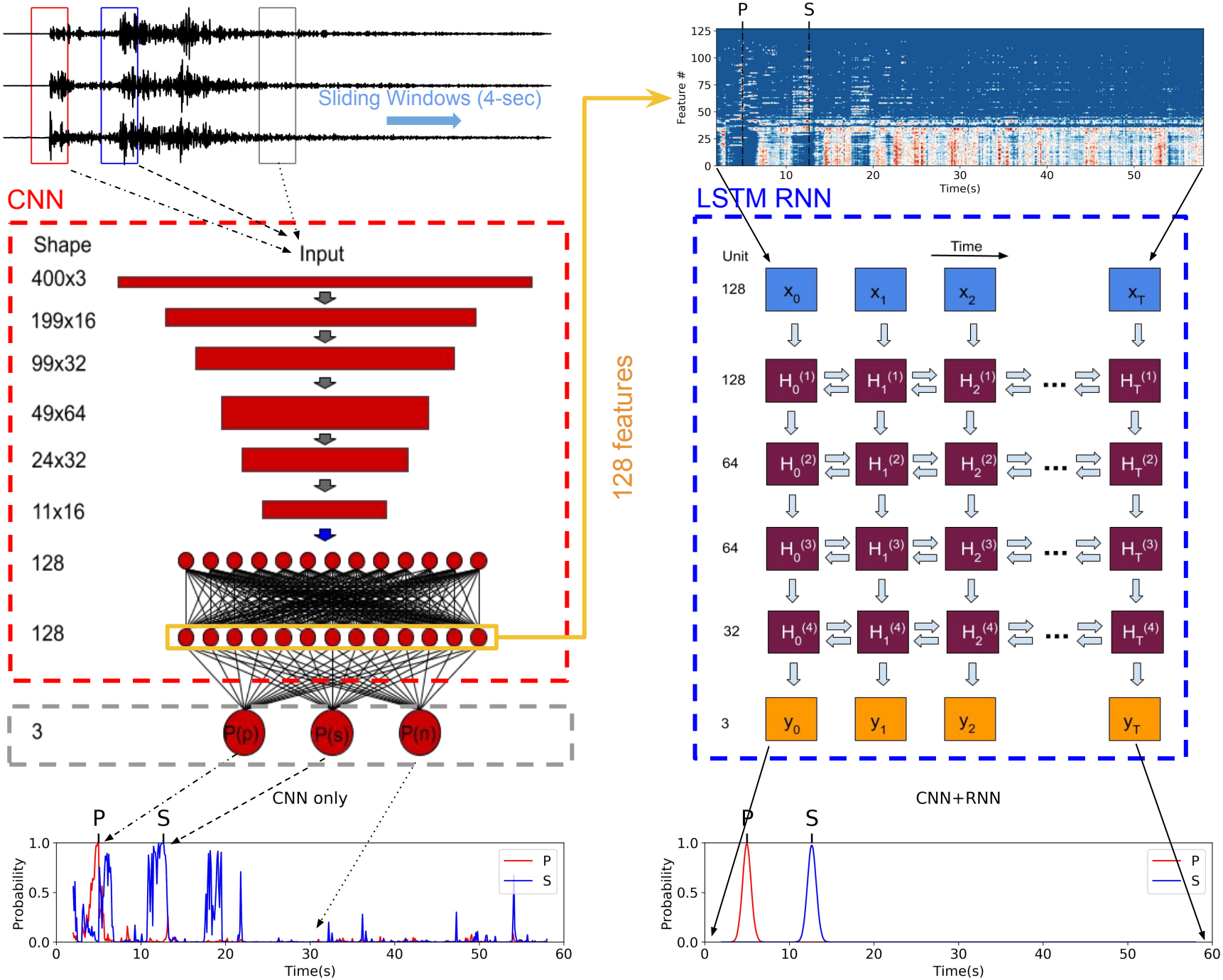
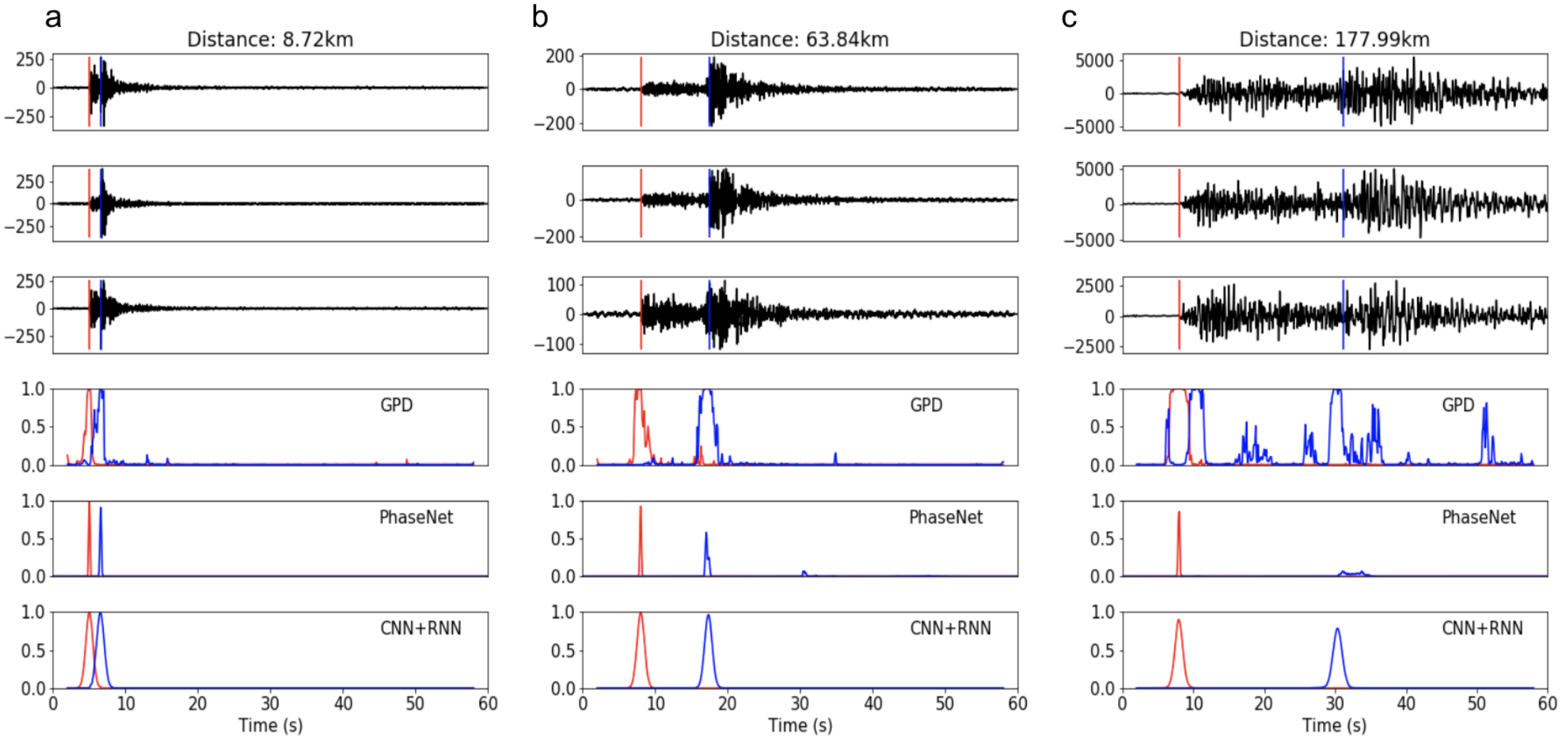
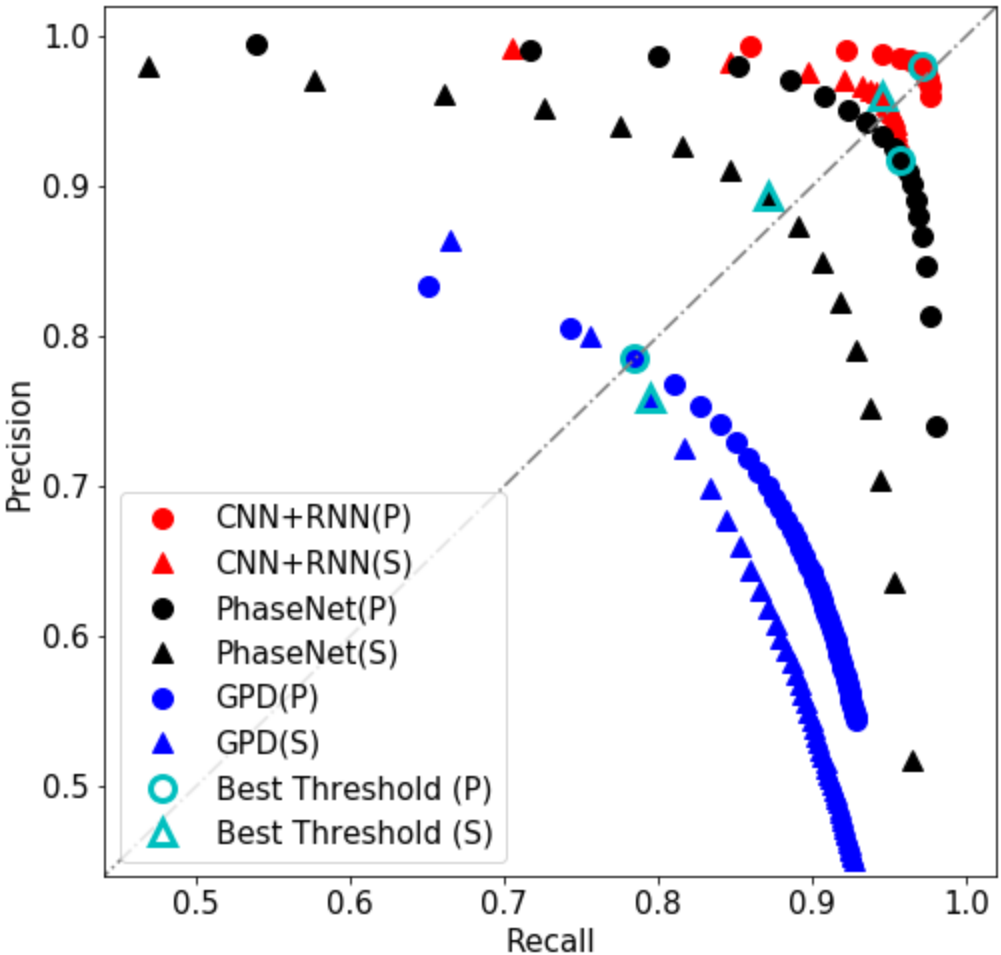
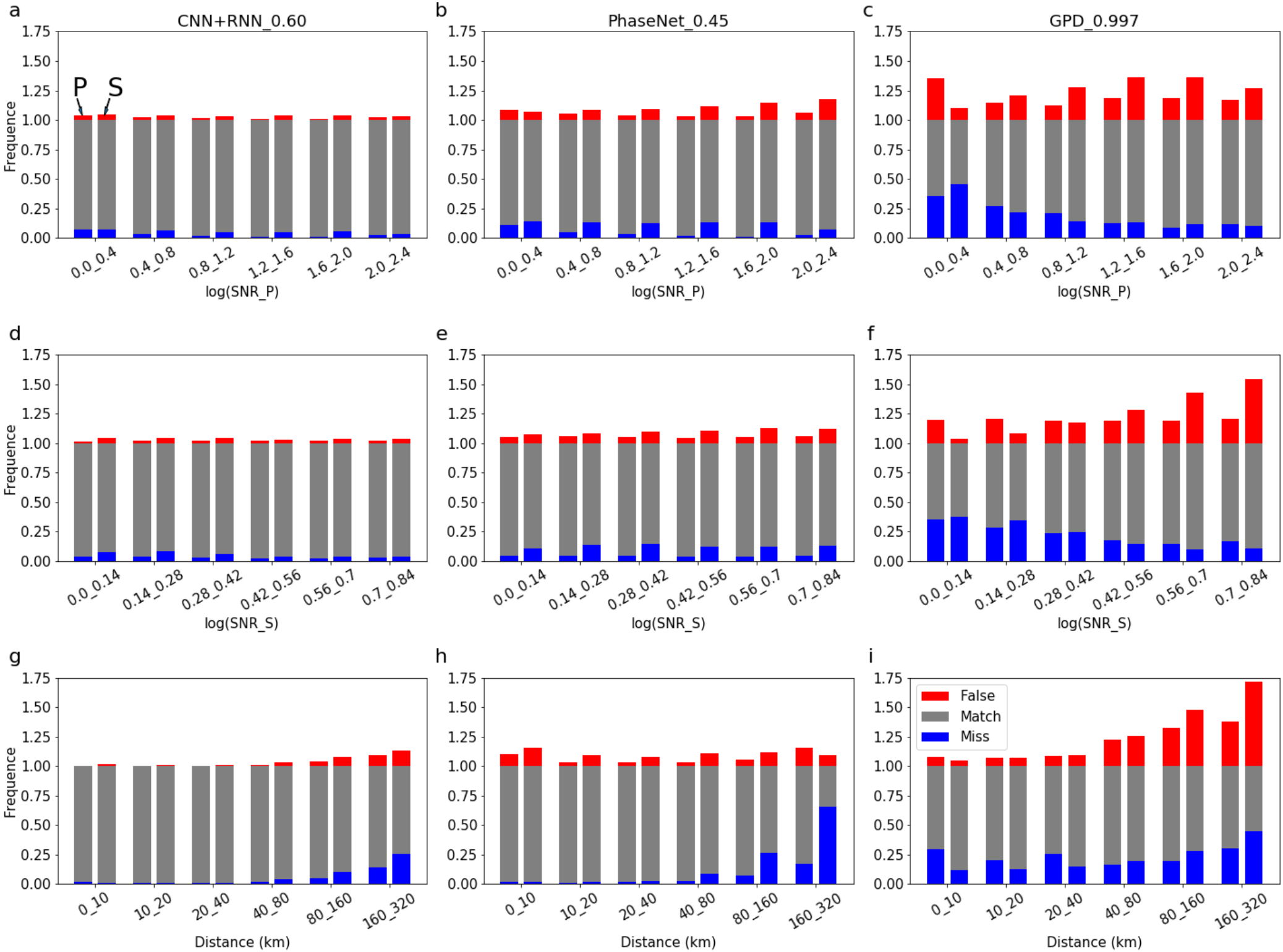
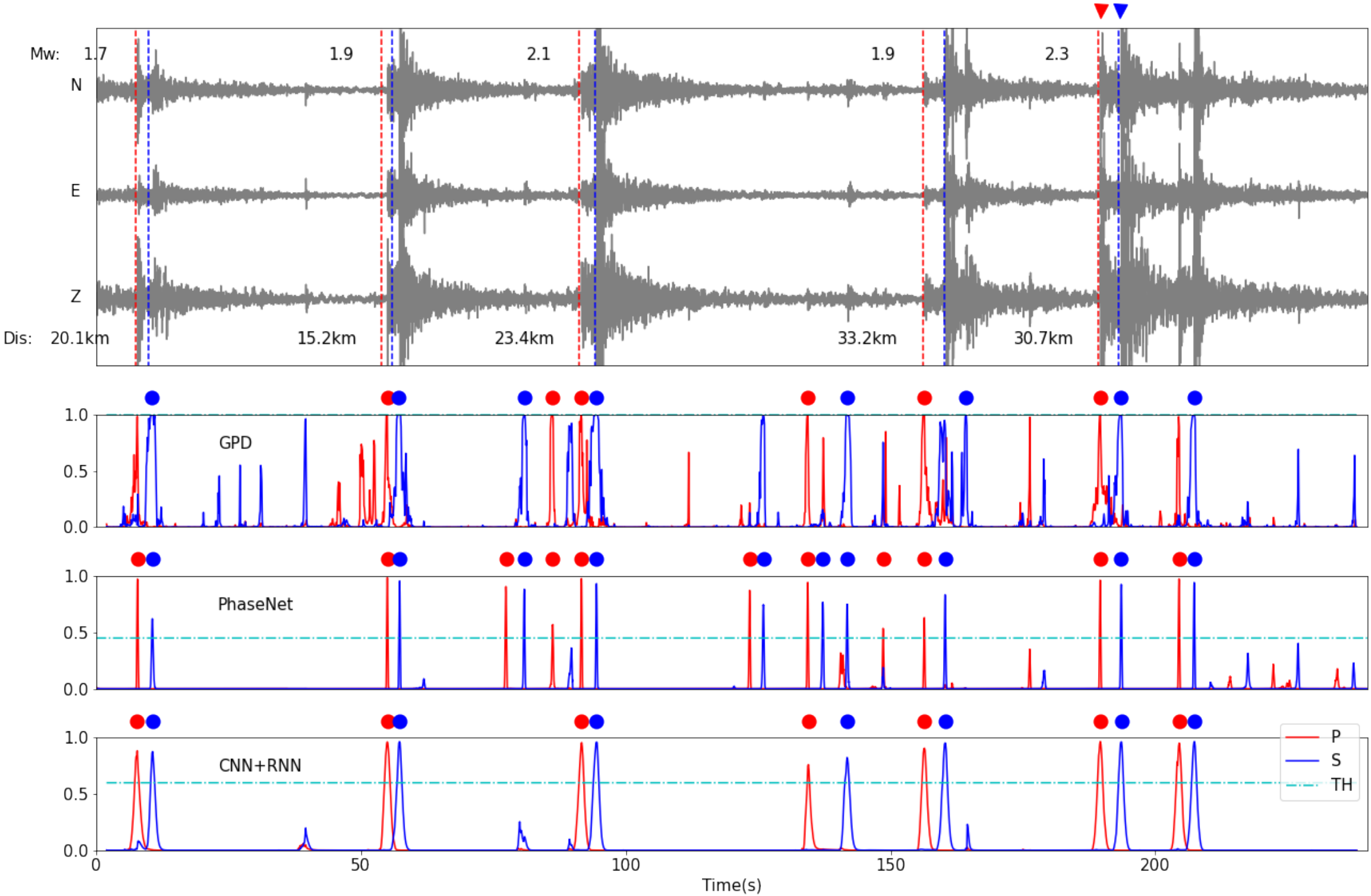
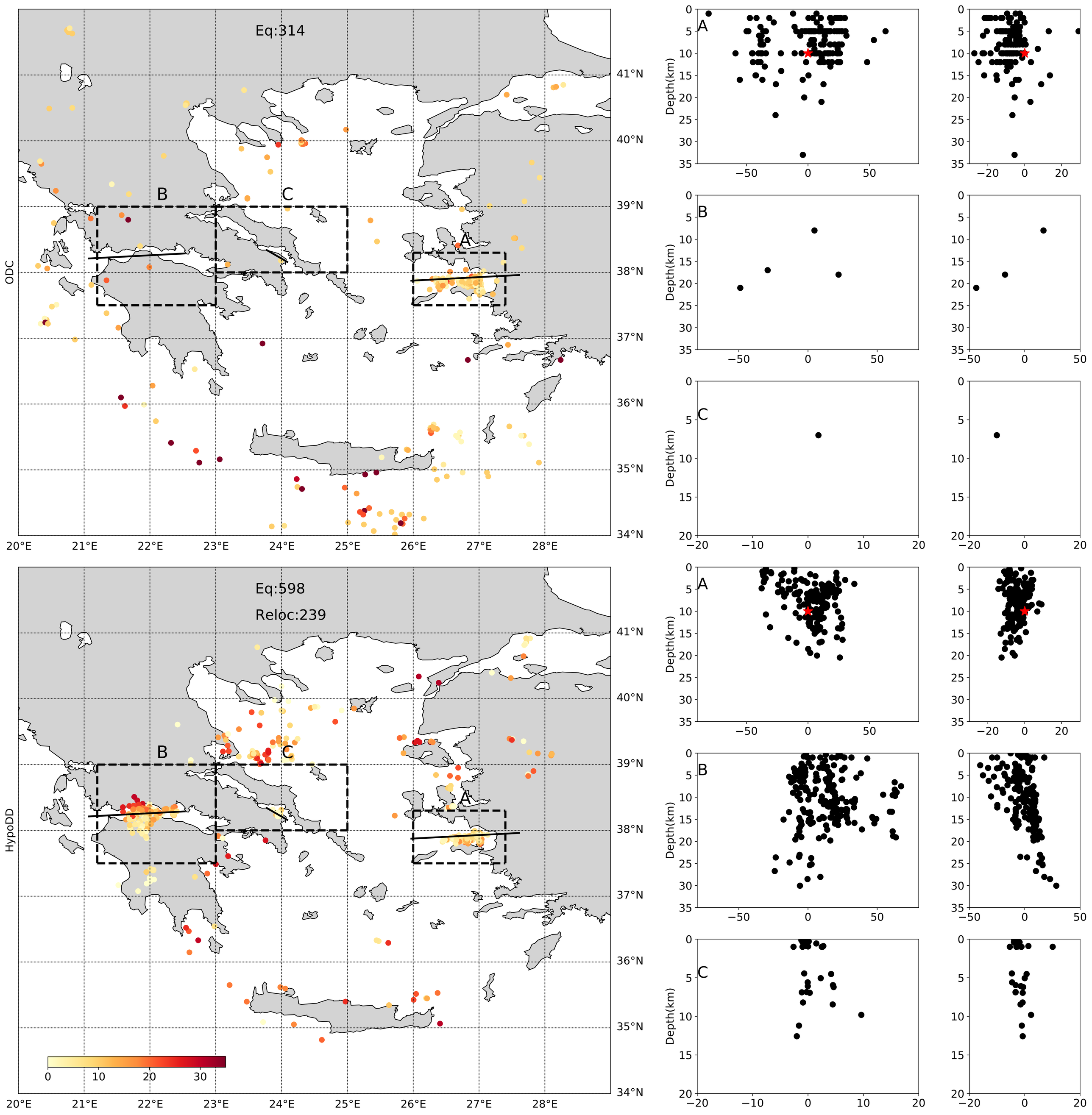

Sliding Windows (4-sec)
CNN
Shape
Input
400x3
199x16
99x32
49x64
24x32
11x16
128
128
3
P(p)
P(s)
P(n)
128 features
P
S
Feature #
Time(s)
LSTM RNN
Unit
Time
128
64
64
32
3
CNN only
CNN+RNN
Probability
Time(s)

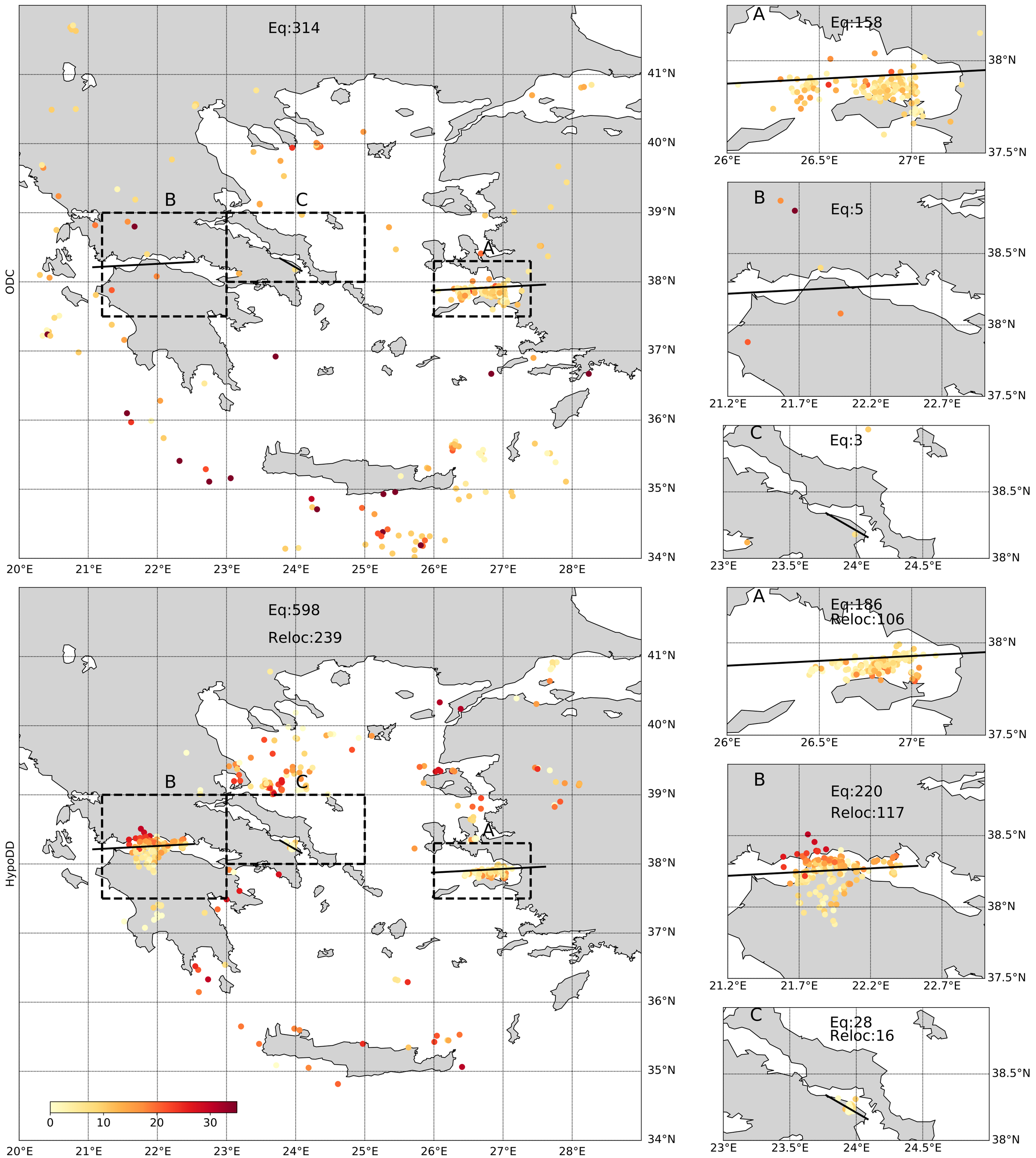

a
Distance: 8.72km
b
Distance: 63.84km
c
Distance: 177.99km
GPD
PhaseNet
CNN+RNN
Time (s)

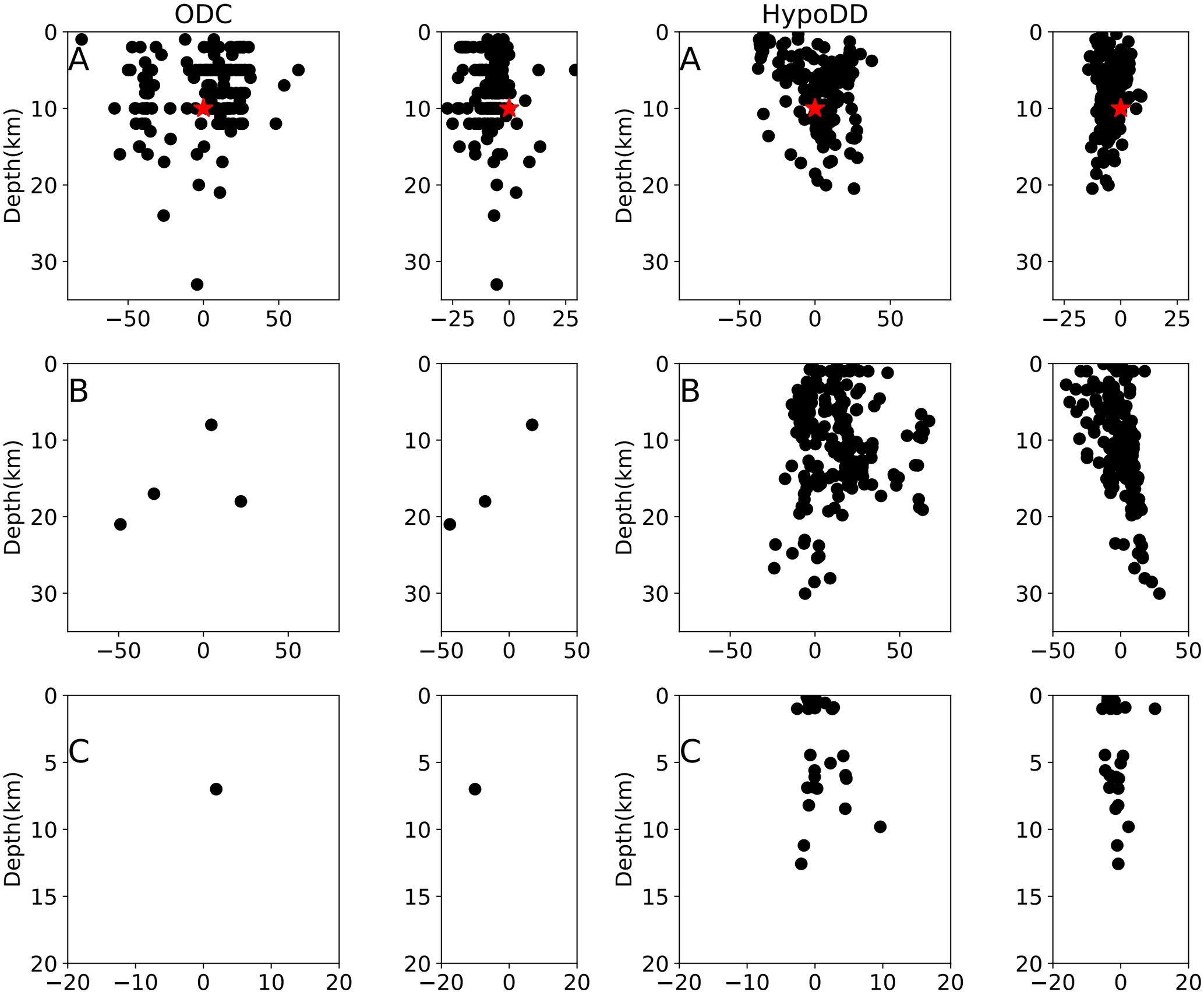

1.0
0.9
0.8
0.7
0.6
0.5
Precision
0.5
0.6
0.7
0.8
0.9
1.0
Recall
CNN+RNN(P)
CNN+RNN(S)
PhaseNet(P)
PhaseNet(S)
GPD(P)
GPD(S)
Best Threshold (P)
Best Threshold (S)


a
CNN+RNN_0.60
b
PhaseNet_0.45
c
GPD_0.997
P
S
Frequence
log(SNR_P)
0.0_0.4
0.4_0.8
0.8_1.2
1.2_1.6
1.6_2.0
2.0_2.4
d
e
f
log(SNR_S)
0.0_0.14
0.14_0.28
0.28_0.42
0.42_0.56
0.56_0.7
0.7_0.84
g
h
i
Distance (km)
0_10
10_20
20_40
40_80
80_160
160_320
False
Match
Miss


Mw:
1.7
1.9
2.1
1.9
2.3
N
E
Z
Dis:
20.1km
15.2km
23.4km
33.2km
30.7km
GPD
PhaseNet
CNN+RNN
P
S
TH
1.0
0.5
0.0
0
50
100
150
200
Time(s)


Eq:314
ODC
Eq:598
Reloc:239
HypoDD
A
B
C
Depth(km)
41°N
40°N
39°N
38°N
37°N
36°N
35°N
34°N
20°E
21°E
22°E
23°E
24°E
25°E
26°E
27°E
28°E
0
10
20
30


Eq:314
ODC
A
B
C
41°N
40°N
39°N
38°N
37°N
36°N
35°N
34°N
20°E
21°E
22°E
23°E
24°E
25°E
26°E
27°E
28°E
Eq:158
38°N
37.5°N
26°E
26.5°E
27°E
Eq:5
38.5°N
38°N
37.5°N
21.2°E
21.7°E
22.2°E
22.7°E
Eq:3
38.5°N
38°N
23°E
23.5°E
24°E
24.5°E
Eq:598
Reloc:239
HypoDD
0
10
20
30
Eq:186
Reloc:106
Eq:220
Reloc:117
Eq:28
Reloc:16


ODC
HypoDD
A
B
C
Depth(km)